"Interactive Dashboard for Laboratory Quality Control Monitoring (2023)"

Goal

The goal of this project is to develop an interactive dashboard to monitor quality control(QC) in clinical laboratory testing throughout the year 2023 by utilizing the **patient laboratory test data**. Laboratories usually rely on routine QC measurements to ensure the accuracy and reliability of test results. These dashboards help lab technicians, quality officers, and analysts identify patterns of deviation, highlighting analytes or devices with frequent violations, and visually track trends in performance over time. By enabling timely detection of out of control measurements, the dashboard supports better decision making and process improvement. The dashboard will be able to evaluate measured values by comparing them against their respective lower and upper control limits, helping to determine if the device falls in the normal range or not. By tracking repeated tests over time ,The dashboard will allow medical professionals to observe increases or decreases in critical values and act proactively. Additionally when the same analysis is conducted across 2 different devices, the system will be able to compare outcomes to assess measurement consistency and device accuracy. The dashboard will generate alerts and suggestions to support clinical decision making. It will also enhance transparency by providing drill-down capabilities into specific tests, timeframes, and instruments offering a streamlined, user-friendly interface to improve test result reliability, ensure equipment performance, and support quality control in medical laboratories.

Design Considerations / Selected Visualizations

The design of the dashboard is driven by clarity, interactivity, and effectiveness in communicating key metrics. Filters can be added to view the dashboard based on detailed test logs by device, by specific time range, and by test type. Color coded indicators can be added to highlight values that fall outside control limits. Flexible time range options, such as daily, weekly, monthly, and quarterly views, can be included. Non-numeric attributes are derived to indicate the current status of the measured value whether it is lower than the limit, within range, or higher than the limit. An export option can be provided for offline audits. Key Performance Indicators (KPIs) can be calculated to evaluate the efficiency and accuracy of devices.

Different distinct visualizations were selected to address different analytical needs.Visualisation techniques such as **Time Series Line Charts**, **Bar Charts**, **KPI Cards**, **Radial Device Charts**, and **Control Funnel Plots** are used. The **Time Series Line charts** help track measured values over time revealing trends, spikes or declines in the patients results or test stability. **Bar Chart** provides a clear summary of frequency distributions, such as how often each device falls out of control range. **KPI cards** offer a glance of metrics like percentage of in range tests and probability of a disease enabling quick assessment. **Control Funnel Plot** displays deviations of measured values from their control ranges over

time, using visual cues like color and thresholds to instantly flag problematic results. The **Radial Device Chart** offers a circular representation of analyzer performance, allowing users to quickly compare how often each device stays within control limits across multiple tests. A **Calendar Heatmap** provides a temporal overview, showing how violations are distributed throughout the year, revealing patterns such as seasonal trends or recurring issues on specific days. Together, these visualizations create a comprehensive, intuitive tool for monitoring and improving laboratory quality.

Mockups

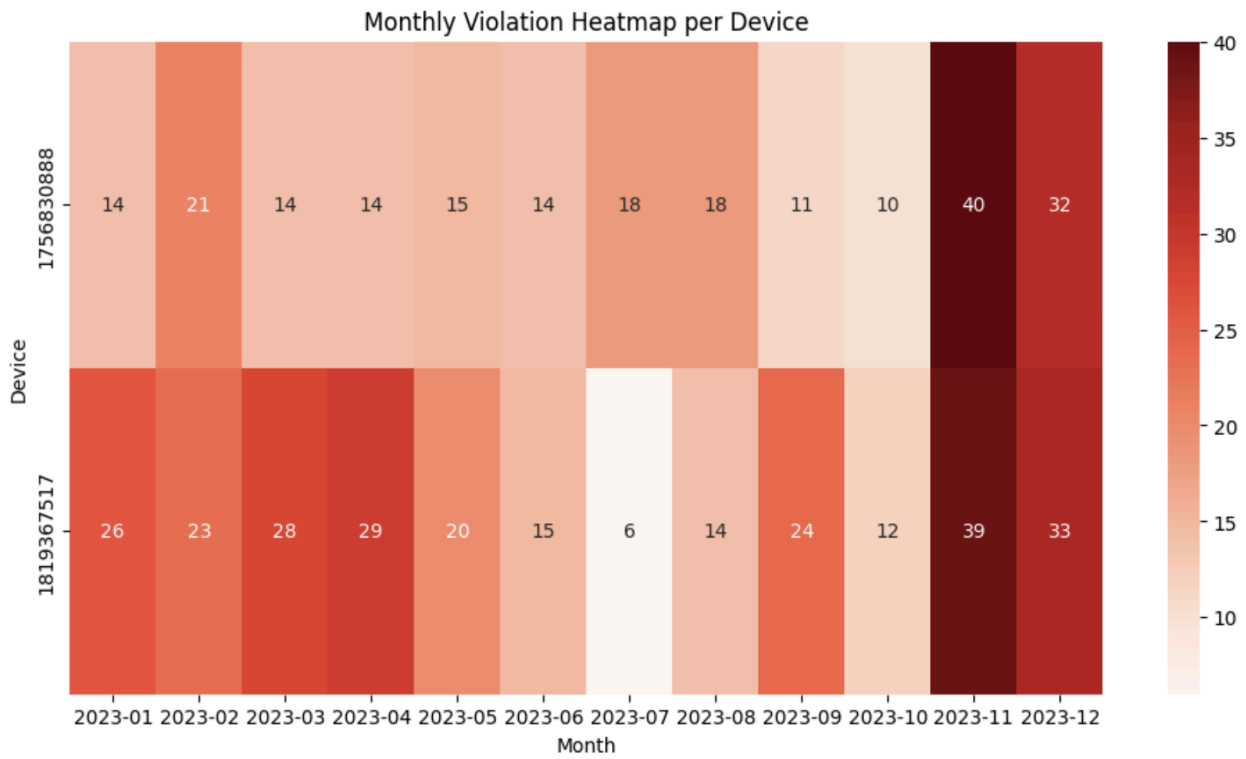


Figure 1: Heatmap plotted with rows representing Device ID and columns representing specific month and the number inside each cell indicates how many violations occurred for that device in that month. Deeper red, mean the higher number of violations whereas Lighter colors (or white) indicate fewer or no violations.

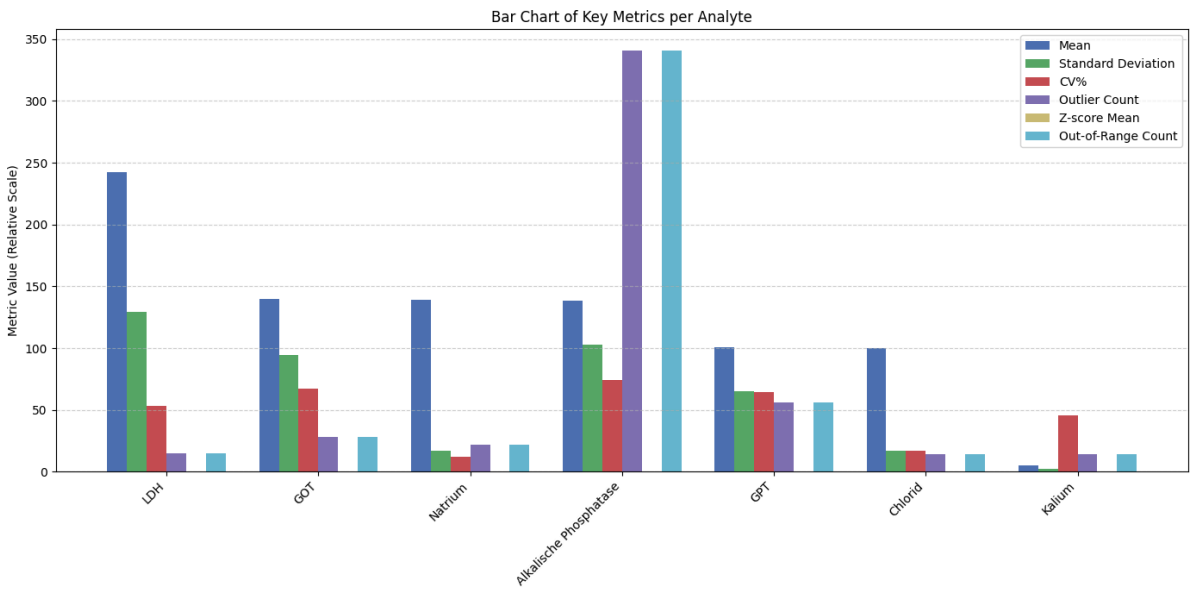


Figure 2 : This figure compares six key quality metrics Mean, Standard Deviation, CV%, Outlier Count, Z-score Mean, and Out-of-Range Count for the top 10 analytes.This visualization helps identify analytes with high variability, frequent outliers, or poor precision. It supports quick performance comparison and decision-making for quality control in the lab.

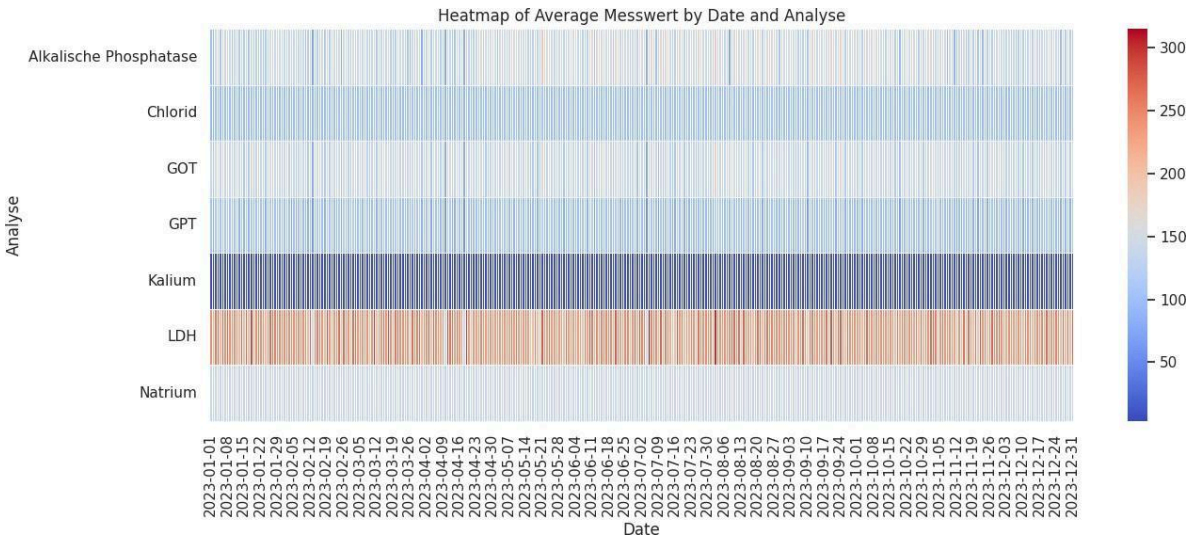


Figure 3 : This displays the average daily measurement values (`Messwert`) for various clinical tests in 2023. Each row represents a different analyte, and color intensity reflects value magnitude blue for low, red for high. The plot reveals stable trends across tests with no major anomalies over time.

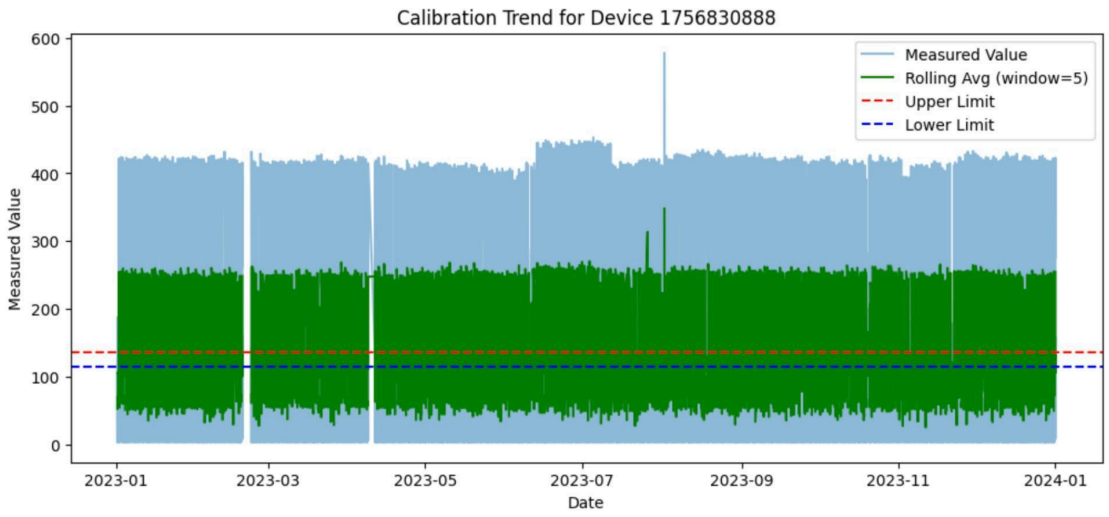


Figure 4 : This compares the measured values (blue) with a 5-point rolling average (green) and shows the acceptable calibration range through upper and lower limits (red dashed lines).Large fluctuations beyond these limits may require further investigation into the device's calibration.

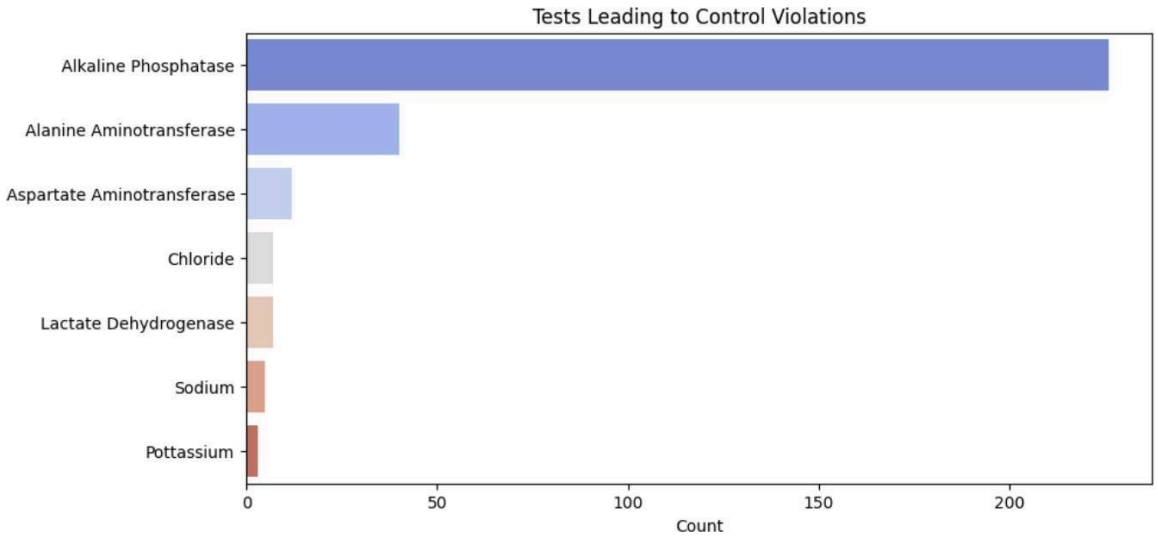


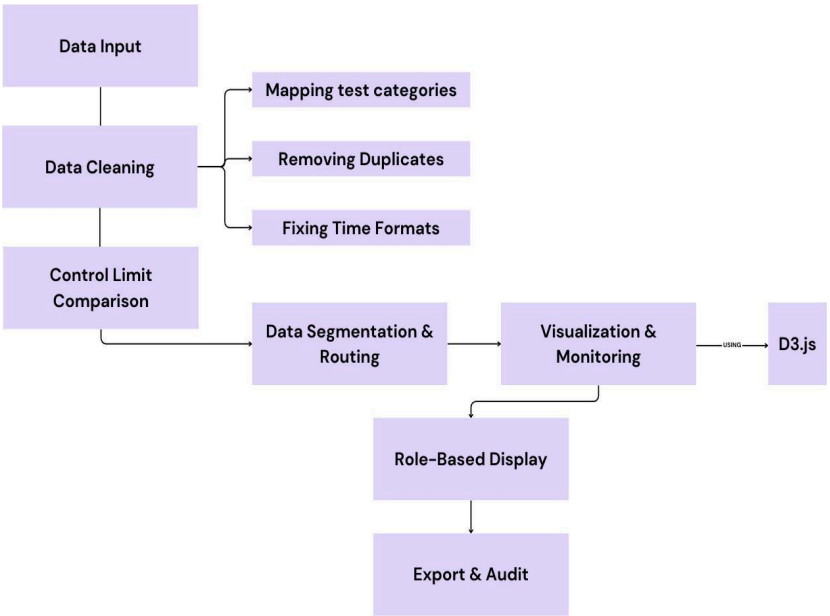
Figure 5 : This bar chart displays the frequency of control violations across various tests. Alkaline Phosphatase stands out with a significantly higher count compared to the others. This highlights areas of concern for better test control management.

The five visualizations work together to provide a comprehensive overview of the laboratory's performance, supporting proactive quality control measures. Figure 1 (Heatmap) tracks device-specific violations across the year, helping identify devices with recurrent issues and seasonal trends, enabling technicians to address consistent problems. Figure 2 (Key Quality Metrics Comparison) compares metrics like Mean, Standard Deviation, CV%, Outlier Count, Z-score Mean, and Out-of-Range Count across analytes, highlighting tests with high variability or poor precision, thus improving test consistency. Figure 3 (Average

Daily Measurement Values) offers an overview of daily test measurements with color intensity to spot anomalies or inconsistencies, ensuring tests remain stable and controlled. Figure 4 (Measured Values vs. Rolling Average) compares individual test measurements with a rolling average and control limits to monitor device calibration stability, helping detect and address calibration issues before they impact results. Lastly, Figure 5 (Frequency of Control Violations) summarizes control violation frequencies, pinpointing tests or analytes with frequent violations and enabling focused corrective actions to improve overall testing accuracy. Together, these visualizations allow lab managers and quality officers to monitor, detect, and correct issues in real time, ensuring reliable test results and efficient quality control.

Implementations & External Libraries

The system receives raw test data, followed by cleaning steps such as fixing time formats,, identifying blanks, and mapping test categories. Next, measured values are compared to their respective control limits, and the status (For example: Low, Normal, High) is updated accordingly. This processed data is then segmented and routed to dashboards, where it is visualized using D3.js. Role-based filters are applied, and the dashboard is published and tested to validate the results. Finally, an export option is provided for external quality audits or offline analysis.



The tasks above outline a structured approach to visualizing and analyzing lab data. Key visualizations provide a comprehensive view of laboratory performance, combining time-based analysis (Figures 1 and 3) with detailed device and test insights (Figures 2, 4, and 5). This allows laboratory staff to quickly identify issues and take corrective actions to maintain quality and accuracy.

In real-life scenarios, the dashboard helps clinical labs handle large volumes of tests by offering a real-time overview of test performance, ensuring compliance with control limits and supporting regulatory audits.

Future updates will introduce dynamic features like drill-downs, interactive filters, and real-time updates, enhancing user interactivity, providing deeper insights, and improving the overall experience for better quality control in clinical laboratories.